Theory of Computation 1-7 Revlew of Mathematical Theory

Let us discuss each one by one -

i) Negation

This proposition is also called NOT proposition. It is denoted by 1 or ~ or having

horizontal bar on letter.

If S is a proposition then S defines NOT s.The truth table for this connective is

S	1s
T	F
F	T

Conjunction (AND

If A and B are two propositions then conjunction of A and B is denoted by A^B. The

truth table for this connective is

A	B	A^B
T	T	T
T.	F	F

A	B	A^B
F	T	F
F	F	F

iii) Disjunction OR

If A and B are two propositions then disjunction of A and B is denoted by AvB.The

truth table for this connective is

A	B	AvB
T	T	T
T	F	T.
F	T	T
F	F	F

iv) Implication

Jf A and B are two propositions then if A then B is a proposition.It can be denoted

by ABorA=B.

The truth table for this connective is

A	B	AB
T	T	T
T	F	F
F	T	T
F	F	T

For example: Find truth value for following proposition.

"If an apple is not green then it must be red.

Let, A denotes An apple is not green" and B denotes It must be redThe statement

A is false and B is true. Hence the proposition is true.

vIf and only if

If A and B are two propositions then A if and only if B.It can be denoted as AB

The truth table will be

A	B	AB
T	T	T
T	F	F

A	B	AB
F	T	F
F	F	T

For example: I will go to the school if and only if it is not raining.Here statement A

is I will go to the school. The statement B is It is not raining. Thus the resultant

proposition is false.

1.5 Relations

Relaonship is a major aspect between two objects, even this is true in our real life.

One ofect can be related with the other object by a some relation.Then those two objects

forma pair based on this certain relationship.

Definition The relation R is a collction for the set S which represents the pair of

elements.

For example (a b is in R.We can represent their relation as a R b. The first

pair is chosen from a set called range

1.9 Review of Mathematical Theory

Properties of Relations

A relation R on set S is

1. Reflexive if iRi for all i in S.
2. Irreflexive if iRi is false for all i in S.
3. Transitive if iRj and jRk imply iRk
4. Symmetric if iRj implies jRi.
5. Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive.

For example: if A=a, b| then

reflexive relation R can be =(a.ab, b

irreflexive relation R can be =|ab1

transitive relation R can be=ab)b, aa, a

symmetric relation R can be=|a, bb, a

asymmetric relation R can be =a.b)

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive,

over some set S.

Suppose R is a set of relations and S is the set of elements.

For example: S is the set of lines in a plane and R is the relation of lines intersecting

to each other.

A relation is said to be antisymmetric if a, b) e R and b, a e R then a =a

Example 1.5: Determine whether R is equivalence relation or not where

A=0121R=00.10.11)2, 2)2, 1

Solution: The R is reflexive because 0, 0).1, 1).2, 2)e R.

Where as R is not symmetric because (0, 1) is not in R whereas (1, 0) is in R.

Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given, il may not be reflexive or transitive

By adding some pairs we make the relation as reflexive or transitive.

For exampleLet 1ab)bc)a, abb